

The pathwidth complex: twelve problems, one width, and a solver proved sound

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Abstract

Draft abstract, to be rewritten with Section 1. Linhares and Yanasse (2002) listed twelve problems from operations research, VLSI design and graph theory as equal, up to one, to the minimum number of open stacks, and gave no proofs. We state each problem precisely and prove what is true, in Lean: seven of the twelve, and two variants, equal pathwidth or pathwidth plus one exactly; two lie in bands of width one and two; two are false as stated. A bibliometric census shows that the literature has settled on one of the twelve names, pathwidth, and that the communities rarely cite one another. We then read the exact open-stacks search of Chu and Stuckey (2009) as a pathwidth algorithm. Two of its published dominance theorems are false; a 14-vertex graph refutes the first, and the same graph refutes its restatement in Chu's thesis. The repaired rule is an instance of Tamaki's commitment lemma and costs one bipartite matching per candidate. The whole repaired search is proved sound in Lean. Run as the default in both of our solvers, it has re-refuted 112 of the 115 corpus values that rested on the published rules, changing none, and it emits certificates that an independent checker verifies. The equivalences make one certified width an answer to every problem in the exact core, and we release 17,714 isomorphism classes from twenty benchmark collections, 16,087 with a certified pathwidth and a witness layout.

1 Introduction

Placeholder. Section 1 waits on the owner's PhD thesis (Linhares, *Industrial pattern sequencing problems: some complexity results and new local search models*, 2001), which is not yet held. It is to record how the twelve problems of Table 1 of Linhares and Yanasse (2002) were collected, what the thesis proved and what it only cited, and why the table appeared with no proofs and no graph (`paper2/plan.md`, section 1). Kashiwabara and Fujisawa (1979), Table 1's interval-thickness reference, is still missing.

Contributions (draft list, for the introduction to argue).

- A census of how much each of the twelve names is used, after removing the search hits that use the words in other senses; pathwidth has more relevant works than the other eleven names together, three times over (Section 2).
- Each problem of Table 1 stated as an instance and a question, with the graph it lives on and its exact relation to pathwidth, every relation proved in Lean; two of Table 1's rows are false as stated (Section 3).

*Draft of 3 October 2026, assembled from the project's working documents (`paper2/`). Every number regenerates by the command named beside it in those documents; the theorems are checked in Lean 4 with Mathlib.

- The exact open-stacks search of [Chu and Stuckey \(2009\)](#) read as a vertex-separation search: two of its published dominance theorems are false, as are their restatements by [Chu \(2011\)](#) and [Fink \(2012\)](#); the repair, its cost, and a Lean proof that the repaired search is sound; and a proof object for its refutations with an independent checker (Section 4).
- A dataset of 17,714 isomorphism classes from twenty collections, 16,087 of them with a certified pathwidth and a witness layout, which through Section 3 answers every problem in Table 1’s exact core (Section 4.7).

Notation. Graphs are finite and simple, $G = (V, E)$, $n = |V|$. A *layout* is a bijection $L : V \rightarrow \{1, \dots, n\}$. For a 0/1 matrix M with rows R (customers, nets, piece types) and columns C (patterns, gates, products), the *MOSP graph* G_M has vertex set R , with $r \sim r'$ when some column has a 1 in both rows; each column becomes a clique on its rows. For an order π of the columns, row r is *active* at position j if it has 1s in columns x, y with $\pi(x) \leq j \leq \pi(y)$.

2 One complex, twelve names

Table 1 of [Linhares and Yanasse \(2002\)](#) gathers twelve problems from three disciplines. From operations research it takes the minimization of open stacks (MOSP); from VLSI design, gate matrix layout, PLA folding and one-dimensional logic; from graph theory, pathwidth, vertex separation, interval thickness, node search, edge search, narrowness, split bandwidth and edge separation. Section 3 shows which of them are in fact one quantity. This section asks a prior question: when the literature studies that quantity, which name does it use? The answer gives the paper its title. Taken together, the problems are one object seen from three disciplines, and the name the literature has converged on is pathwidth.

2.1 Method

We measured how much each name is used with OpenAlex ([Priem et al., 2022](#)), on 29 September 2026. For each of the twelve problems we searched titles and abstracts for its name and its common variants, and restricted the search to the four fields in which the problems are studied: Computer Science, Mathematics, Engineering and Decision Sciences. Without that restriction several names are ordinary words: “narrowness” returns over a million works, and “edge separation” returns aerodynamics.

Inside those fields a search hit is still not a work that uses the name. OpenAlex’s search stems words and ignores punctuation, so a quoted phrase matches more than the phrase. “Narrowness” matches “narrowing” in term rewriting; “node searching” matches look-ups in peer-to-peer networks; “split bandwidth” matches radar interferometry and 5G spectrum assignment; “path-width” matches the widths of roads, tool paths and datapaths. We therefore fetched all 2,271 hits with their titles, abstracts, topics and venues, and kept a work only if it is about the problem in the sense of Table 1. For the eleven smaller names the 700 hits were judged one by one. For pathwidth, the 1,571 hits were filtered by a rule: keep a work that uses the one-word spelling, that OpenAlex places in a theory subfield, or that mentions graphs, treewidth or minors and not roads, vehicles, trajectories or datapaths. We checked the rule by reading samples of what it kept and what it dropped.

The check changed the counts and two of the conclusions. Of the 2,271 hits, 1,593 are relevant: 1,213 for pathwidth and 380 for the other eleven names.¹ Three names have no relevant work at all. The two search-game names keep 27% and 40% of their hits. MOSP moves from seventh place to fourth.

¹OpenAlex returned one pathwidth record twice. The counts are of records, as the figures draw them; there are 1,212 distinct relevant pathwidth works, 1,592 in all.

Table 2.1: Works that use each problem’s name in the sense of Table 1, and citations of the paper Table 1 cites for it. *Relevant*: hits kept by the check of Section 2.1. *Citations*: OpenAlex `cited_by_count`, 29 September 2026. Kashiwabara and Fujisawa (1979) has no OpenAlex record. ^a1,212 distinct works.

Problem	Discipline	Relevant	Hits	Table 1 reference	Citations
Path-width	graph theory	1,213 ^a	1,571	Kinnersley (1992)	214
Gate matrix layout	VLSI	110	125	Möhring (1990); Wing et al. (1985)	134 / 89
PLA folding	VLSI	68	74	Möhring (1990)	134
MOSP	operations research	58	60	Yanasse (1997a); Fink and Voß (1999)	75 / 71
Vertex separation	graph theory	55	100	Kinnersley (1992)	214
Edge search	graph theory	38	95	Kirousis and Papadimitriou (1986)	293
Node search	graph theory	34	127	Kirousis and Papadimitriou (1985)	124
One-dimensional logic	VLSI	11	18	Ohtsuki et al. (1979)	108
Interval thickness	graph theory	6	10	Kashiwabara and Fujisawa (1979)	–
Narrowness	graph theory	0	35	Kornai and Tuza (1992)	44
Split bandwidth	graph theory	0	35	Fomin (1998)	19
Edge separation	graph theory	0	21	Lengauer (1981)	77
All twelve		1,593	2,271		

The counts are lower bounds. A work can study a problem without naming it in its abstract, OpenAlex has few abstracts before about 1990, and its citation counts run below Google Scholar’s. The labels were made in one model-assisted session, record a yes or no per work and not the reason, and have not been checked by a second labeller. As a second reading we applied a mechanical rule, the literal phrase with a context term and no exclusion term. It agrees with the labels on 543 of the 700 labelled works (78%). Under both readings pathwidth is more than three times the other eleven names together (3.2 and 3.6 times), the three empty names stay empty apart from one 5G false positive, and the search games lose most of their hits. MOSP’s rank is the one conclusion that depends on the reading: fourth under the labels, fifth under the rule. The labels, the rule, the cached responses and the code that redraws every figure are in the repository.²

2.2 Pathwidth has absorbed the family

Table 2.1 and Figure 2.1 give the counts. At 1,213 relevant works pathwidth is 21 times MOSP and more than three times the other eleven names put together (380). The axis of Figure 2.1 is linear on purpose: the claim is about the size of that gap. Anything one would want to know about algorithms for this class of problems was most likely published under “pathwidth”.

Three of the twelve names are not in use at all, and two more barely. Narrowness, split bandwidth and edge separation have no relevant work; every hit was about something else. Split bandwidth is Fomin’s own coinage in the paper Table 1 cites (Fomin, 1998), and nobody adopted it. Interval thickness (6 works) and one-dimensional logic (11) are nearly as rare. MOSP is one of the better-used names, at 58 relevant works, behind pathwidth and the two VLSI layout names.

Citation counts do not track name usage. Kirousis and Papadimitriou (1986) is the most-cited paper in the table (293 citations), while only 38 works use “edge search” in its sense: the paper is cited as a

²paper2/relevance.py, paper2/trends.py, paper2/section2_figures.py; regenerate with `python -m paper2.section2_figures, offline, from the caches of 29 September 2026.`

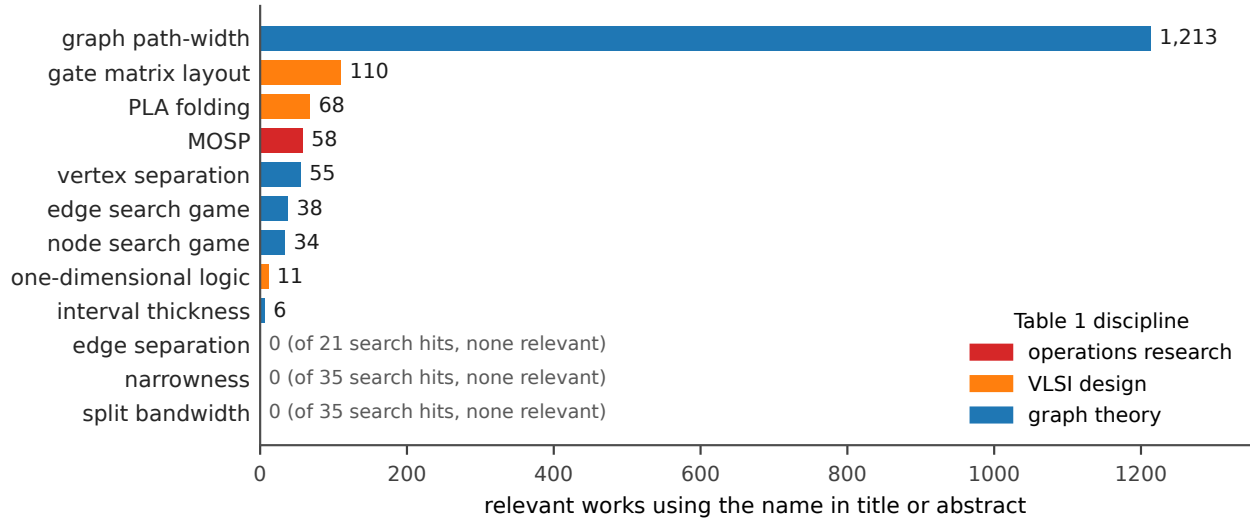


Figure 2.1: Works that use each problem’s name, in title or abstract, in the sense of Linhares and Yanasse (2002, Table 1). OpenAlex, fields Computer Science, Mathematics, Engineering and Decision Sciences, all years, fetched 29 September 2026. Only works judged relevant are counted (Section 2.1). Colour gives the discipline that Table 1 assigns the problem. OpenAlex returned one pathwidth record twice, so pathwidth’s 1,213 are 1,212 distinct works.

foundational result on graph searching, not because people work on the edge search game under that name. The split is starker for Ohtsuki et al. (1979): 108 citations against 11 works using “one-dimensional logic”.

2.3 Three generations

Figure 2.2 shows when each name was alive. The shading is a rate, so that the growth of publishing as a whole does not make every row rise, and each row is scaled to its own peak, so that a name with six works reads as clearly as one with a thousand. The VLSI names belong to the 1980s: PLA folding and gate matrix layout peak in 1980–89 and fall to almost nothing after 2010, and one-dimensional logic has no relevant work after 2009. The graph-searching names peak later and fade. MOSP is the operations-research generation: it first appears in the 1990s, peaks in 2005–14, and is still active. Only pathwidth is growing. Its rate has roughly doubled since the 1990s, from about 15 to 27 works per million, and its 311 relevant works in 2020–24 are more than the whole history of any other name. Rows with a few works per period move by whole cells on one paper, and should be read as anecdote rather than trend.

2.4 Communities that rarely cite each other

Figure 2.3 asks whether the three communities read each other. Works that cite one Table 1 paper ring that paper; works that cite several are pulled inside the circle. Of the 844 citing works, 269 cite two or more Table 1 papers, but 195 of those stay inside one discipline. 74 span two or more, and 63 of those join graph theory to VLSI. The operations-research side is an island: only six works cite both a MOSP paper and a graph-theory paper. Two are by the authors of Table 1; one is MOSP work on graph properties; three are pathwidth or vertex-separation work from computer science. The fields follow the disciplines. Works citing the MOSP papers are mostly in Engineering (86 of 135), and works citing the graph-theory papers mostly in Computer Science (417 of 507).

Two consequences shape the rest of the paper. First, a result on any of the twelve problems should be stated, and searched for, as a result on pathwidth; Section 4 does this for an exact MOSP algorithm and finds both a known theorem it had rediscovered and a published error that the pathwidth literature’s commitment

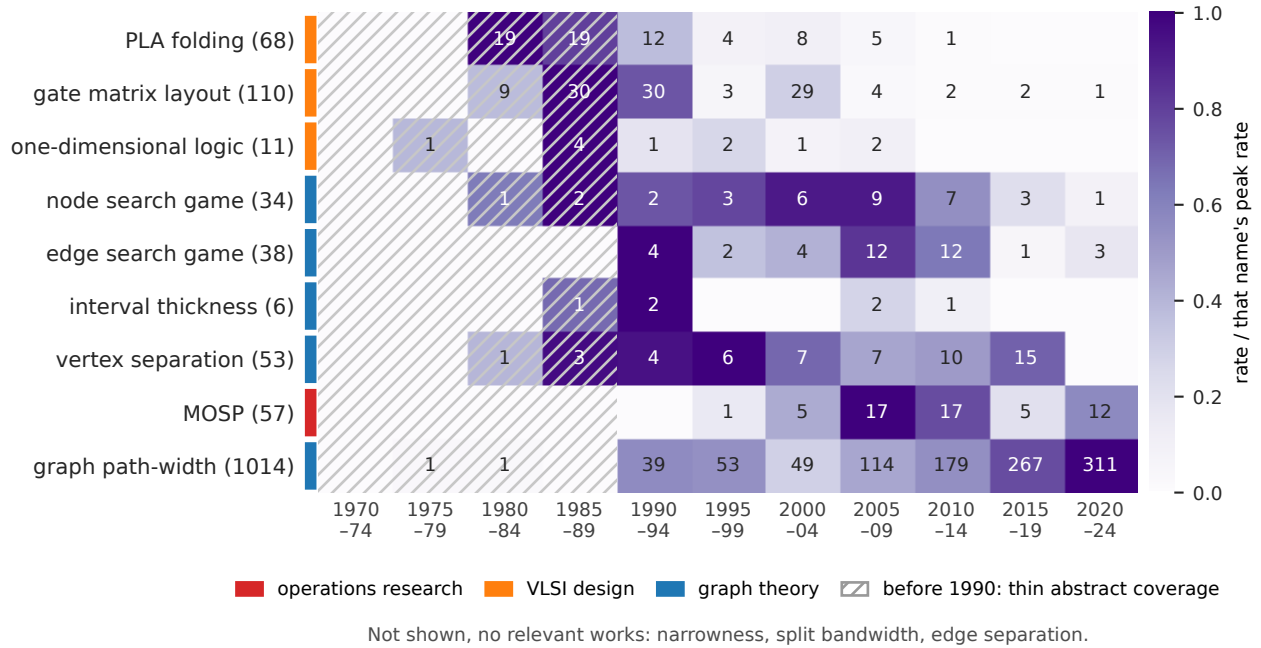


Figure 2.2: When each name was in use. The shade is the name’s rate, relevant works per million works in the same four fields in each five-year period, as a fraction of that name’s own peak rate. The numbers are raw counts. Rows are ordered by the period of their peak. Counts cover 1970–2024, since 2025–26 is not fully indexed; this is why pathwidth’s row totals 1,014 here, against 1,213 in Figure 2.1. Before 1990 (hatched) OpenAlex has few abstracts, so the early VLSI names are undercounted there.

lemma rules out. Second, a benchmark for any problem in the exact core is a pathwidth benchmark, which is how Section 4.7 builds its dataset.

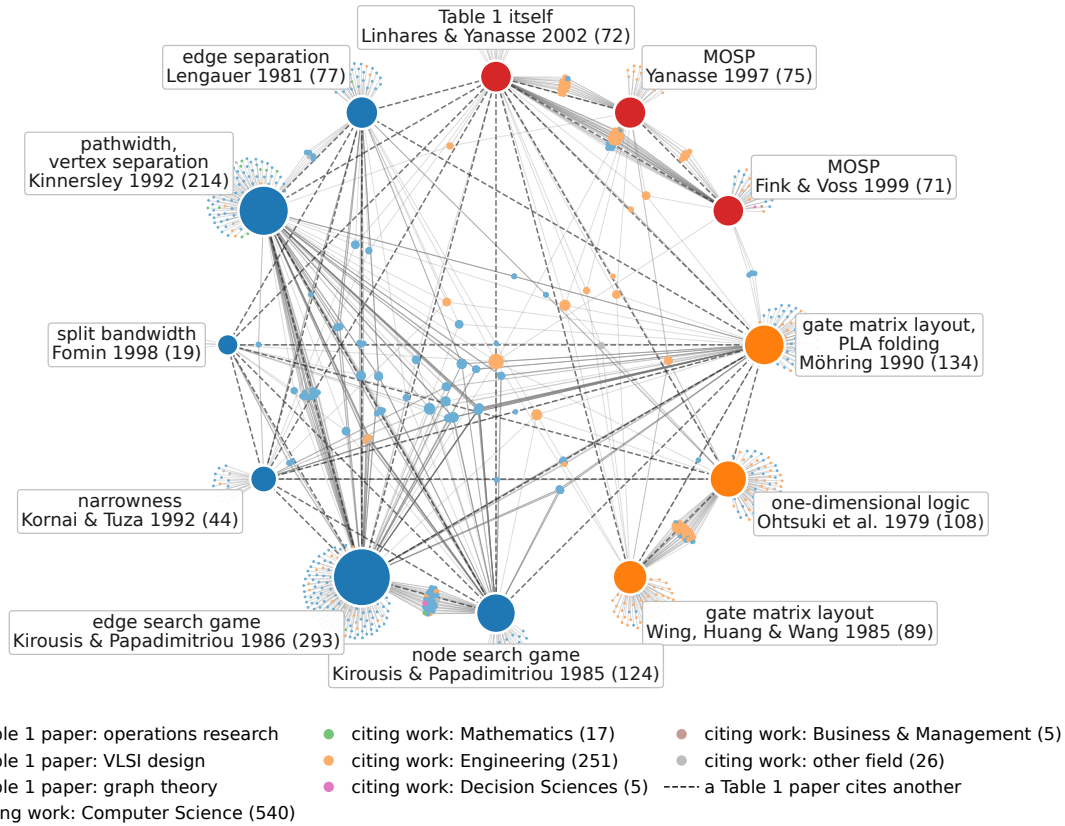


Figure 2.3: The twelve Table 1 papers that OpenAlex indexes, on a circle, sized by citations (in brackets) and coloured by discipline, and the 844 works that cite them, coloured by OpenAlex field. 269 of the 844 cite two or more Table 1 papers. 74 of them cite papers of two or more disciplines, counting the eleven problem papers and not Linhares and Yanasse (2002) itself, and 63 of those 74 join graph theory to VLSI. Only 6 cite both a MOSP paper (Yanasse, 1997a; Fink and Voß, 1999; Linhares and Yanasse, 2002) and a graph-theory paper. Kashiwabara and Fujisawa (1979) has no OpenAlex record.

3 The equivalences, proved

Linhares and Yanasse (2002, p. 1764) introduce Table 1 as “a set of problems that consist of, given input Π , compute a function $f(\Pi)$ that is either equal to the number of open stacks or closely related to it (plus or minus one)”. The table cites where each problem was studied. It gives no proofs and no graph. Two readings of “plus or minus one” must be kept apart. Under the *literal* reading, f lies within one of Z , the number of open stacks of the corresponding instance, on every input. Under the *fixed-offset* reading, $f = Z + c$ for one constant $c \in \{-1, 0, 1\}$ on every input. Only the second makes computing one problem the same as computing another.

This section states each problem as an instance and a question, gives the relation to pathwidth that is true, and says where it is proved. Every relation stated below is a theorem of our Lean development unless it says otherwise. All the theorems named depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`.³ Table 3.1 summarises the result.

³Printed by `lake env lean paper2/axiom_check.lean`. The development has one sorry, a conjecture about lower bounds that no result of this paper uses.

3.1 The problems

Each problem below is a decision problem with an instance and an integer k ; its value is the least k for which the answer is yes.

Pathwidth $\text{pw}(G)$. Is there a sequence of bags $X_1, \dots, X_\ell \subseteq V$ covering V , with every edge inside some bag, $X_i \cap X_\ell \subseteq X_j$ whenever $i \leq j \leq \ell$, and $|X_i| \leq k + 1$? (Kinnersley, 1992, p. 346).

Vertex separation $\text{vs}(G)$. Is there a layout L such that for every $i < n$ at most k vertices u with $L(u) \leq i$ have a neighbour v with $L(v) > i$? (Kinnersley, 1992, p. 346).

MOSP $Z(M)$. Given a 0/1 matrix M of customers by patterns, is there an order of the patterns with at most k customers active at every position? (Linhares and Yanasse, 2002, eqs. (1)–(2)).

Gate matrix layout $t(M)$. Given a net–gate matrix, is there an order of the gates and an assignment of nets to k tracks such that nets active at a common position get different tracks? (Möhring, 1990, p. 18); (Wing et al., 1985, Problem 1).

One-dimensional logic. Given gates, nets and the nets each gate connects, is there an order of the gates whose net intervals have an interval graph of clique number at most k ? Its *connection graph* H joins two nets that share a gate. A *boundary* variant pins two gates to the ends (Ohtsuki et al., 1979, §§II–IV).

PLA folding $\text{pla}(M)$. As gate matrix layout, with at most two nets per track. With no such bound (*multiple folding*) it is gate matrix layout itself (Möhring, 1990, pp. 25, 36).

Interval thickness $\theta(G)$. Is there an interval supergraph of G on the same vertices with clique number at most k ? (Kirkousis and Papadimitriou, 1985, p. 182); (Möhring, 1990, p. 31).

Node search $\text{ns}(G)$. Can searchers, placed on and removed from vertices, clear every edge with at most k on the graph at once, where an edge is cleared when both ends are guarded and a clear edge is recontaminated by a searcher-free path to a contaminated one? $\text{mns}(G)$ is the same over strategies that never recontaminate (Kirkousis and Papadimitriou, 1985, p. 181); (Kirkousis and Papadimitriou, 1986, §2).

Edge search $\text{es}(G)$. The same game with a third move, sliding a searcher along an edge, which clears that edge; $\text{pes}(G)$ for progressive strategies (Kirkousis and Papadimitriou, 1986, p. 208); (Ellis et al., 1994, p. 53).

Narrowness $\nu(G)$. Is there an order $v_1, \dots, v_n$ such that the following process never holds more than k vertices: at step i add v_i , then remove every v_j , $j \leq i$, with no neighbour v_ℓ , $\ell > i$? (Kornai and Tuza, 1992, §2).

Split bandwidth $\text{sb}(G)$. Is there a graph obtained from G by repeatedly splitting a vertex into an edge, its neighbours partitioned between the two ends, with bandwidth at most k ? (Fomin, 1998, §3.2).

Edge separation. Table 1 cites Lengauer (1981), which supports three readings: cutwidth $\text{cw}(G)$, the least over layouts of the most edges crossing a gap (p. 468); modified cutwidth $\text{mcw}(G)$, counting only edges that pass strictly over a vertex (Definition 6); and Lengauer’s vertex separator game $\text{VSG}(G)$, the least K such that the vertices can be pebbled in an order that leaves at most K unpebbled vertices adjacent to pebbled ones (p. 467).

Table 3.1: Table 1 of Linhares and Yanasse (2002), settled. *Relation*: what is true and proved in Lean, on the graph the problem lives on (the MOSP graph or net graph for the matrix problems, the input graph otherwise). *Status*: *exact*, a fixed offset; *band*, an inequality whose ends both occur; *false*, Table 1’s claim fails, with a counterexample family proved in Lean.

Problem	Proved in	Relation	Status
MOSP	Yanasse (1997b); Fellows and Langston (1989); Linhares and Yanasse (2002)	$Z(M) = \text{pw}(G_M) + 1$ (M has a 1)	exact
Gate matrix layout	Möhring (1990)	$t(M) = Z(M) = \text{pw}(G_M) + 1$	exact
One-dim. logic	Ohtsuki et al. (1979)	$\text{tracks} = \theta(H) = \text{pw}(H) + 1$; boundary variant $\geq \text{pw}(H) + 1$	exact
PLA folding	Möhring (1990)	$\max(\text{pw} + 1, \lceil N /2 \rceil) \leq \text{pla}$, gap unbounded; multiple folding $= \text{pw} + 1$	false; variant exact
Interval thickness	Möhring (1990); Kirousis and Papadimitriou (1985)	$\theta = \text{pw} + 1$ ($V \neq \emptyset$)	exact
Node search	Kirousis and Papadimitriou (1985, 1986); Bienstock and Seymour (1991)	$\text{ns} = \text{mns} = \text{vs} + 1 = \text{pw} + 1 = \theta$ ($E \neq \emptyset$)	exact
Edge search	Ellis et al. (1994); Kirousis and Papadimitriou (1986)	$\text{vs} \leq \text{es} \leq \text{vs} + 2$, also for pes	band
Narrowness	Kornai and Tuza (1992)	$v = \text{pw} + 1$ ($V \neq \emptyset$)	exact
Split bandwidth	Fomin (1998)	$\text{pw} \leq \text{sb} \leq \text{pw} + 1$	band
Path-width	Kinnersley (1992)	the reference	definition
Edge separation	Lengauer (1981)	cw, mcw unbounded against pw (stars); $\text{VSG} = \max(1, \text{vs})$	false; VSG exact
Vertex separation	Kinnersley (1992)	$\text{vs} = \text{pw}$	exact

3.2 What is true

Two lemmas carry most of the exact rows. For a path decomposition and a vertex v , let $\text{first}(v)$ and $\text{last}(v)$ be the least and greatest indices of bags containing v ; the decomposition’s interval property says $v \in X_i$ exactly when $\text{first}(v) \leq i \leq \text{last}(v)$.

Lemma 3.1 (Helly for intervals). *Pairwise intersecting intervals of a line have a common point.*

Proof. Let a be the largest left end. Any two intervals meet, so every right end is at least every left end, and in particular at least a . So a lies in all of them. $\square$

Lemma 3.2 (cliques sit in a bag). *In a path decomposition of G , every clique of G lies inside one bag.*⁴

Proof. The bags containing a vertex u of the clique form the interval $[\text{first}(u), \text{last}(u)]$. Two vertices of the clique are adjacent, so their edge lies in a bag and their intervals meet. By Lemma 3.1 all the intervals share an index i , and the clique lies in X_i . $\square$

Theorem 3.3 (vertex separation is pathwidth). *For every graph, $\text{vs}(G) = \text{pw}(G)$.*⁵

⁴Lean: PathDecomposition.exists_bag_of_isClique.

⁵Lean: vertexSeparation_eq_pathwidth.

This is Theorem 3.1 of [Kinnersley \(1992\)](#). From a layout L of separation k , the bags $X_i = V_L(i-1) \cup \{L^{-1}(i)\}$, where $V_L(i)$ is the set the definition counts, form a decomposition of width k . Conversely, ordering the vertices by first gives a layout in which everything counted at a cut lies in the bag where the next vertex first appears, beside that vertex.

Theorem 3.4 (MOSP is pathwidth plus one). *If M has at least one 1, then $Z(M) = \text{pw}(G_M) + 1$.*⁶

Proof. ($\leq$) Take a decomposition of G_M of width k . The rows of each column form a clique, so by Lemma 3.2 they lie in a bag $X_{b(c)}$. Order the columns by $b(c)$. If row r is active at the position of column c , it has columns x, y with $b(x) \leq b(c) \leq b(y)$, and $r \in X_{b(x)} \cap X_{b(y)}$, so the interval property gives $r \in X_{b(c)}$. At most $k+1$ rows are active there.

($\geq$) Take an order with at most Z active rows at every position. The active sets, with a singleton bag for each row with no 1, form a path decomposition: rows sharing a column are both active at its position, and a row is active on an interval of positions. Its width is $Z-1$, using $Z \geq 1$. $\square$

Read as a cutting problem, Theorem 3.4 is Proposition 5 of [Yanasse \(1997b\)](#), which turns each pattern into a clique on its piece types; read as a layout problem, it is Theorem 7 of [Fellows and Langston \(1989\)](#), with the column-expansion Lemma 4.1 of [Fellows and Langston \(1987\)](#); and [Linhares and Yanasse \(2002, Prop. 2\)](#) identify MOSP with gate matrix layout. The proof above needs the Helly property and the interval property, not Hall's theorem. It needs no assumption that the instance is reduced.

The graph matters. The pattern graph, with the patterns as vertices joined when they share a customer, gives no bound in either direction: one pattern shared by four customers who each have a private pattern has $Z = 4$, while its pattern graph is the star $K_{1,4}$, of pathwidth 1.⁷ A single customer with many patterns errs the other way.

The other exact rows attach to Theorems 3.3 and 3.4.

Theorem 3.5 (the exact core).

1. Gate matrix layout is MOSP on the same matrix: if M has a 1, $t(M) = Z(M) = \text{pw}(G_M) + 1$. For a fixed gate order the fewest tracks equals the most nets active at one position, by the left-edge algorithm.⁸
2. Interval thickness: if $V \neq \emptyset$, $\theta(G) = \text{pw}(G) + 1$ ([Möhring, 1990, Prop. 3.5](#)).⁹
3. One-dimensional logic: if every net lies on a gate, the fewest tracks is $\theta(H)$, and if some gate connects a net it is $\text{pw}(H) + 1$ ([Ohtsuki et al., 1979, Thm 3](#)).¹⁰
4. Narrowness: if $V \neq \emptyset$, $v(G) = \text{pw}(G) + 1$, and for each order σ , $v(\sigma) = \text{vs}(\sigma^{\text{rev}}) + 1$ ([Kornai and Tuza, 1992, Prop. 3.1](#)).¹¹
5. Node search: for every graph $\text{ns}(G) = \text{mns}(G)$, and if $E \neq \emptyset$, $\text{ns}(G) = \text{vs}(G) + 1 = \text{pw}(G) + 1 = \theta(G)$ ([Kirousis and Papadimitriou, 1985, Thm](#)); ([Kirousis and Papadimitriou, 1986, Thms 2.3, 4.1](#)).¹²
6. Multiple folding: the fewest tracks is $t(M) = \text{pw}(G_M) + 1$ ([Möhring, 1990, Thm 3.14](#)).¹³

⁶Lean: `MOSPInstance.mospValue_eq_pathwidth_add_one`.

⁷Lean: `star_mospValue, star_pathwidth_agreementGraph, star_refutes_pattern_graph_bound`.

⁸Lean: `NetGateMatrix.tracks_eq_mospValue, NetGateMatrix.tracks_eq_pathwidth_add_one`.

⁹Lean: `intervalThickness_eq_pathwidth_add_one`.

¹⁰Lean: `LogicArray.tracks_eq_intervalThickness, LogicArray.tracks_eq_pathwidth_add_one`.

¹¹Lean: `narrowness_eq_pathwidth_add_one`.

¹²Lean: `nodeSearchMonotonicity, nodeSearch_eq_vertexSeparation_add_one, nodeSearch_chain`.

¹³Lean: `NetGateMatrix.foldTracks_eq_pathwidth_add_one`.

7. *Lengauer’s vertex separator game*: $\text{VSG}(G) = \max(1, \text{vs}(G))$ (Lengauer, 1981, Thm 4).¹⁴

The Lean proof that node search is monotone follows the crusade argument of Bienstock and Seymour (1991), not the reduction to edge search of LaPaugh (1993) on which Kirousis and Papadimitriou (1986) rely.

Theorem 3.6 (bands). *For every graph, $\text{pw}(G) \leq \text{sb}(G) \leq \text{pw}(G) + 1$ (Fomin, 1998, Thm 8), and $\text{vs}(G) \leq \text{es}(G) \leq \text{vs}(G) + 2$, and the same for pes (Ellis et al., 1994, Thm 2.1).*¹⁵

Every end of both bands occurs. K_2 has $\text{sb} = \text{pw} = 1$ and $\text{es} = \text{vs} = 1$; the star $K_{1,3}$ has $\text{pw} = 1$ and $\text{sb} = 2$, and $\text{vs} = 1$ with $\text{es} = 2$; and $K_{3,3}$ has $\text{vs} = 3$ with $\text{es} = 5$. These values come from a brute-force check of every quantity from its own definition, not from Lean.¹⁶ Fomin assumes a connected graph with at least two vertices; the Lean proof needs neither. The equality $\text{es} = \text{pes}$ (LaPaugh, 1993) is stated in Lean as a named proposition and used by no row.

Theorem 3.7 (two rows are false).

1. *PLA folding: for a matrix with a 1, $\max(\text{pw}(G_M) + 1, \lceil |N|/2 \rceil) \leq \text{pla}(M)$, and the gap to $\text{pw} + 1$ is unbounded. The 5×5 identity needs 3 tracks while $\text{pw}(G_M) + 1 = 1$, and the path P_n as a matrix, one gate per edge, has $t = 2$ and $\text{pla} \geq \lceil n/2 \rceil$, so the gap is unbounded on connected instances too (Möhring, 1990, Prop. 3.15).*¹⁷
2. *Edge separation read as cutwidth: $\text{pw} \leq \text{cw}$ always, but $\text{cw}(K_{1,n}) \geq n/2$ and $\text{mcw}(K_{1,n}) \geq n/2 - 1$ while $\text{pw}(K_{1,n}) = 1$; so neither cutwidth reading is within a constant of pathwidth.*¹⁸

Table 1’s PLA claim holds for multiple folding (Theorem 3.5). The edge-separation row is also mis-attributed: what Lengauer (1981) proves exactly is the vertex separator game, which is Table 1’s vertex separation row.

3.3 Edge cases and a gap in a published proof

Three of the published equalities fail on degenerate inputs, and each failure is proved in Lean. On a nonempty graph with no edges, $\text{ns} = 0$ while $\text{vs} + 1 = \theta = 1$, so both equalities of Kirousis and Papadimitriou (1985, 1986) need an edge.¹⁹ For a matrix with no 1s, $Z = 0$ while $\text{pw} + 1 = 1$. Lengauer’s Theorem 4 fails at $K = 0$ on an edgeless graph.²⁰ Möhring (1990, Thm 3.9), $t = \text{ns}$, fails literally on the 1×1 matrix $[1]$.²¹

The published proof of $\text{vs} \leq \text{ns} - 1$ (Kirousis and Papadimitriou, 1986, Thm 4.1) has a gap. It orders the vertices by when they first receive a searcher and claims that every vertex counted at the worst cut still holds a searcher, “since the vertices in D_{i_0} are joined to vertices not yet cleared” (p. 217). A strategy may place a searcher and remove it again before clearing anything, which recontaminates nothing. On $K_{1,3}$, placing and removing a searcher on each leaf before guarding the centre is an optimal recontamination-free strategy for

¹⁴Lean: `vsg_eq_max`.

¹⁵Lean: `pathwidth_le_splitBandwidth_le_pathwidth_add_one, vertexSeparation_le_edgeSearch_le_add_two, vertexSeparation_le_progressiveEdgeSearch_le_add_two`.

¹⁶python -m paper2.complex_check; $\text{sb}(K_{1,3}) \geq 2$ is also argued by hand: splits of a tree are trees and keep at least three leaves.

¹⁷Lean: `plaTracks_idMatrix_five, plaTracks_idMatrix_unbounded, plaTracks_pathMatrix_unbounded`.

¹⁸Lean: `cutwidth_unbounded, modCutwidth_unbounded`.

¹⁹Lean: `nodeSearch_ne_intervalThickness_of_edgeless, monotoneNodeSearch_ne_vertexSeparation_add_one_of_edgeless`.

²⁰Lean: `isPositiveVSG_triangleGraph_counterexample`.

²¹Lean: `tracks_ne_monotoneNodeSearch_one`.

which the claim fails.²² The theorem is true. The repair orders the vertices by clearing time, as our Lean proof does, or uses the paper’s own normal form, its Corollary 2.4.

3.4 A thirteenth member

Progressive black-white pebbling, which is not in Table 1, belongs to the exact core. [Lengauer \(1981, Thm 3\)](#) gives $vs(G) \leq K \iff pbw(G_d) \leq K + 2$ on the dag G_d built from G ; his Theorem 2 gives $pbw(D) = vs(D_u) + 1$ on a dag D , where D_u is the MOSP graph of the matrix whose column v is v with its predecessors; and the minimum progressive pebbling number of [Kirousis and Papadimitriou \(1986, Thm 3.1\)](#) equals $pw + 1$. All three are proved in Lean, each with its hypothesis: Lengauer’s Theorem 2 in its instance form fails at $K = 1$ on a dag with no arcs, and $mpb = ns$ fails on edgeless graphs.²³

Summary. Seven of Table 1’s twelve problems are pathwidth, or pathwidth plus one, exactly: MOSP, gate matrix layout, one-dimensional logic, interval thickness, node search, narrowness and vertex separation, with pathwidth itself as the reference. So are two variants, multiple folding and Lengauer’s vertex separator game. Split bandwidth and edge search are bands of width one and two, which meet the literal reading of Table 1 and not the fixed offset. Simple PLA folding and edge separation are false as stated. One theorem is stated and not proved, LaPaugh’s $es = pes$, and no row needs it. One of Table 1’s references, Kashiwabara and Fujisawa (1979), is not held; its row is proved from [Möhring \(1990\)](#) and [Kirousis and Papadimitriou \(1985\)](#).

4 An exact open-stacks search, read as a pathwidth solver

The best exact method for MOSP on the standard benchmarks is the *customer search* of [Chu and Stuckey \(2009\)](#). It searches over the orders in which the customers’ stacks close, not over product sequences, and prunes with dominance rules, three of them stated as theorems (their Theorems 1, 2 and 3) and a fourth, the subset rule, in their §2. [Chu and Stuckey \(2009\)](#) place MOSP among “graph path-width and gate matrix layout”, citing [Linhares and Yanasse \(2002\)](#), but do not use pathwidth in the search. By Theorem 3.4 it is nevertheless a pathwidth algorithm: any graph is the MOSP graph of the instance with one customer per vertex and one product per edge, whose optimum is pathwidth plus one.

This section reads the search that way, and finds three things. Two of the published theorems are false, and so are their restatements in Chu’s thesis ([Chu, 2011](#)) and, with a stronger premise, in [Fink \(2012\)](#). The repaired first rule is an instance of a known pathwidth theorem, Tamaki’s commitment lemma; the published rule is that lemma with its interior condition dropped. And the whole repaired search is proved sound in Lean. All Lean names in this section are in `lean/MOSPFormalization/Search/`, which has no `sorry` and the standard axioms only.

4.1 The search decides pathwidth

[Chu and Stuckey \(2009, p. 4\)](#) observe that the products matter only through the MOSP graph. So fix a graph $G = (C, E)$ on the customers and an integer $k \geq 0$. For a customer c and a set $T \subseteq C$ of closed customers,

²²Lean: `kirousisPapadimitriou_claim2_false`.

²³Lean: `pbw_lengauerD_eq_pathwidth, pbw_eq_mospValue_pebbleMatrix, mpb_eq_pathwidth_add_one`.

write $N[c]$ for the closed neighbourhood of c , and

$O(T) = \bigcup_{c \in T} N[c]$	the stacks opened once T has closed,
$\text{fin}(X) = \{c : N[c] \subseteq X\}$	the customers free to close when X is open,
$\text{cl}(T) = \text{fin}(O(T))$	the <i>free closure</i> of T ,
$b(T) = O(T) \setminus T $	the stacks open after T has closed,
$o(c, T) = N[c] \setminus O(T), \quad \text{open}(c, T) = o(c, T) $	the stacks closing c would newly open.

A *move* at T closes one customer $c \notin T$, at cost $\text{cost}(T, c) = |O(T \cup \{c\}) \setminus T| = b(T \cup \{c\}) + 1$, the stacks open at that moment counting c ; it is *playable* when $\text{cost}(T, c) \leq k$. Write $P_k(T)$ when some order of $C \setminus T$ has every step playable. On entering a node the search closes every free customer, so it stands only at *states*, sets with $S = \text{cl}(S)$, and the *child* of S by c is $S \cdot c = \text{cl}(S \cup \{c\})$. The search's predicate $\text{Sol}_k(S)$ holds when $S = C$, or when some playable c has $\text{Sol}_k(S \cdot c)$.²⁴ Every node the search visits satisfies the invariant $b(S) \leq k$.

Lemma 4.1 (free moves never hurt). *If $b(T) \leq k$, then $P_k(T) \iff \text{Sol}_k(\text{cl}(T))$.*²⁵

Closing a customer whose neighbourhood is open opens nothing, never makes a later step dearer, and leaves one fewer stack open. The hypothesis is needed: closing the free customers one by one costs $b(T)$ at each step.

Theorem 4.2 (the search decides pathwidth). *For nonempty C , $\text{Sol}_k(\emptyset) \iff \text{pw}(G) + 1 \leq k$; on the MOSP graph of an instance M with a requirement, $\text{Sol}_k(\emptyset) \iff Z(M) \leq k$.*²⁶

Proof. For a closing order $c_1, \dots, c_n$ with prefixes S_i , the i -th step costs $1 + |\partial S_i|$, where ∂X is the set of vertices outside X with a neighbour in X . Read backwards as a layout, the cut separating $C \setminus S_i$ from S_i counts exactly ∂S_i . So the cost of the order is the vertex separation of the reversed layout plus one, and Lemma 4.1 at the root and Theorem 3.3 finish the proof. The Lean statement goes through narrowness (Kornai and Tuza, 1992), which counts the forward order directly. $\square$

So a refutation, $\neg \text{Sol}_k(\emptyset)$, may only mean $\text{pw}(G) > k - 1$. The rest of the section asks whether the pruning rules preserve that meaning.

What a rule must satisfy. At a state S the search has a set Q of *old moves* (Section 4.3). Its candidates are the customers outside $S \cup Q$; the *cost cut* keeps the playable ones, P , in index order, and a *filter* keeps a sublist $L \subseteq P$. The filter is *node-sound* at (S, Q) if, whenever every $q \in Q$ has $\neg \text{Sol}_k(S \cdot q)$, $\text{Sol}_k(S)$ implies $\text{Sol}_k(S \cdot c)$ for some $c \in L$.²⁷ Each rule justifies a discarded move r by a *covering*, a move c with $\text{Sol}_k(S \cdot r) \Rightarrow \text{Sol}_k(S \cdot c)$. A node is sound when every link a rule cites is a true covering and every chain of links from a discarded candidate ends in L or in Q ; the second condition can fail through a cycle even when every link is true. The filter runs the definite move, then the subset rule, then the better move, each once.

4.2 The definite move is false as published

Chu and Stuckey (2009, §3.1, p. 6) state:

Theorem 1. Suppose $S \mathrel{++} [q]$ is playable and $\text{close}(q, S) \geq \text{open}(q, S)$, then if $U' = S \mathrel{++} R$ is a solution, there exists a solution $U = S \mathrel{++} [q] \mathrel{++} R'$,

²⁴Lean: `stepCost, Solvable, SearchSol`.

²⁵Lean: `solvable_iff_searchSol_cl`.

²⁶Lean: `searchSol_empty_iff_pathwidth_add_one_le, searchSol_mospGraph_iff_mospValue_le`.

²⁷Lean: `NodeSoundAt`.

A counterexample to Chu & Stuckey's Theorem 1, at $k = 6$
0 is playable and $\text{close}(0, S) \geq \text{open}(0, S)$, so the rule keeps 0 alone. A solution from S exists
(first moves 1, 3, 4, 6), but none starts with 0.

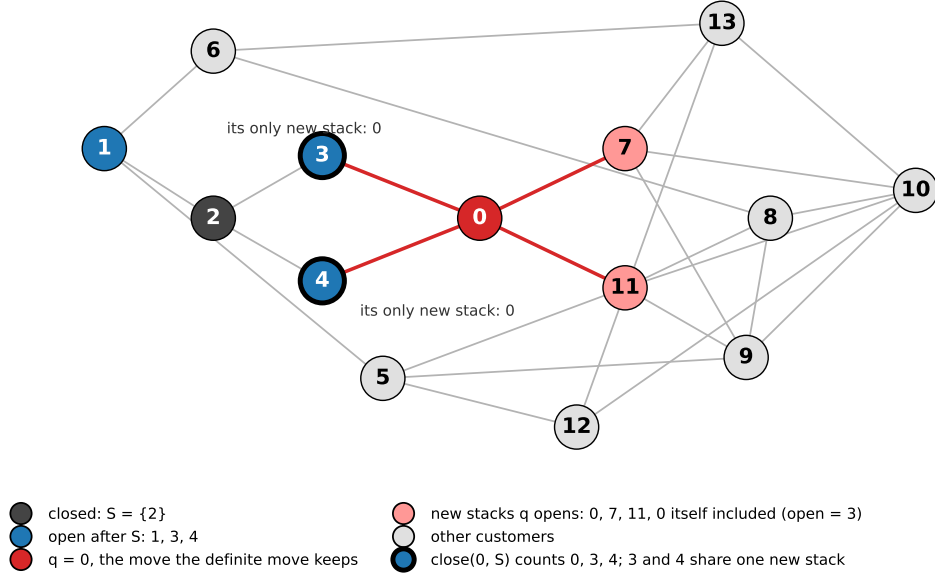


Figure 4.1: The graph of Counterexample 4.4 at the state where the definite move fails. Customer 2 is closed; 1, 3 and 4 are open. Moving 0 opens the new stacks 0, 7 and 11 and closes 0, 3 and 4, so $\text{open} = \text{close} = 3$ and the rule keeps 0 alone. But 3 and 4 have the single new stack 0 between them: closing them first gains two stacks for the price of one, which a solution starting with 0 cannot match. Regenerate with `python -m paper2.counterexample_figure`, which recomputes every labelled fact.

where $\text{close}(q, S)$ counts the customers d with $o(d, S) \subseteq o(q, S)$ (p. 4). The rule keeps q alone. We read d as ranging over customers not yet closed, as the code does; read literally, closed customers count too, which weakens the premise, and both readings are refuted below. Chu (2011, Thm 6.3.6, p. 142) restates the theorem with the same premise and proof, reworded for k stacks. Write $X = S \cdot q$.

Lemma 4.3 (what the premise says). $\text{close}(q, S) = |X \setminus S|$. Hence the premise holds iff $b(X) \leq b(S)$: the child has no more open stacks than the parent.²⁸

Counterexample 4.4. Let G be the graph on customers $0, \dots, 13$ with the 26 edges

0–3, 0–4, 0–7, 0–11, 1–2, 1–5, 1–6, 2–3, 2–4, 5–9, 5–11, 5–12, 6–8,
6–13, 7–9, 7–10, 7–13, 8–9, 8–10, 8–11, 9–10, 9–11, 10–11, 10–12, 10–13, 12–13,

with $S = \{2\}$, $k = 6$ and $q = 0$ (Figure 4.1). Then $O(S) = \{1, 2, 3, 4\}$, $o(0, S) = \{0, 7, 11\}$ and $\text{cost}(S, 0) = 6$, so 0 is playable; the customers with $o(d, S) \subseteq \{0, 7, 11\}$ are 0, 3 and 4, so $\text{close}(0, S) = 3 \geq \text{open}(0, S)$; and 0 comes first in index order, so the rule keeps only 0. A solution exists from S : the order

1, 3, 4, 6, 12, 13, 0, 5, 7, 8, 9, 10, 11

costs at most 6. None exists from the child $X = \{0, 2, 3, 4\}$: only seven sets are reachable from X by moves of cost at most 6, and none is C .²⁹

²⁸Lean: `closeCount_eq, isDefinite_iff`.

²⁹Lean: `definiteMove_counterexample, cexFamily_closed`.

The published implication, as a universal statement, is refuted in Lean under both readings of close, and so is the thesis’s Theorem 6.3.6.³⁰ The smallest graph on which the rule loses the last solution has 14 customers: there is none on any graph to 9 vertices, by exhaustion, and none on the 88,802 connected graphs on 10 vertices with at most 14 edges. The counterexample was found by a search over a family of gadgets built from the failed proof, and minimised by deletion. Fink (2012, pp. 27–28) restates the rule with a stronger premise, counting only dominated customers of smaller index than q . The graph above refutes it under no labelling, but a third twin of 3 and 4 gives a 15-customer graph that does.³¹

Why the published proof fails. The proof moves q to the front of a solution and claims that after each prefix “we have at most $\text{open}(q, S)$ extra stacks open, but at least $\text{close}(q, S)$ extra stacks closed” (p. 6). At a prefix $T \supseteq S$ that does not yet contain q , moving q forward changes the open stacks by $b(T \cup X) - b(T) = |N[q] \setminus O(T)| - |X \setminus T|$. At $T = S$ this is $\text{open} - \text{close} \leq 0$. But customers of X that the solution has *already* closed are in T and no longer count as extra stacks closed, while the stacks q opens may still be new. In the example, a solution may begin with 3, after which 4 is free: at $T = \{2, 3, 4\}$, moving 0 forward costs one more stack. A second, independent slip: even where the conclusion holds, the literal sequence $S \mathrel{++} [q] \mathrel{++} R'$ need not be a solution, because a customer that q frees stays open until its turn; the argument needs the free moves.

The repair. Call q *hereditarily definite* at S when $b(X) \leq b(B)$ for every B with $S \subseteq B \subseteq X$ and $q \notin B$. By Lemma 4.3, the published premise is the case $B = S$. In the counterexample the repair fails at $B = \{2, 3, 4\}$, where $b(B) = 2 < 3 = b(X)$.³²

Lemma 4.5 (open stacks are submodular). *For all $A, B \subseteq C$, $b(A \cup B) + b(A \cap B) \leq b(A) + b(B)$.*³³

Proof. $b(T) = |O(T)| - |T|$, $O(A \cup B) = O(A) \cup O(B)$ and $O(A \cap B) \subseteq O(A) \cap O(B)$; the customers themselves balance exactly. $\square$

Theorem 4.6 (the repaired definite move is sound). *If $q \notin S$ is hereditarily definite at S , then $P_k(S) \Rightarrow P_k(X)$. At a state with the invariant, $\text{Sol}_k(S) \Rightarrow \text{Sol}_k(S \cdot q)$.*³⁴

Proof. Take an order from S of cost at most k with prefixes T_i , in which q is the j -th step, and follow it from X , skipping customers already in X . A step $i < j$ closes c_i at $T_{i-1} \cup X$. With $Z = T_i$ we have $S \subseteq Z \cap X \subseteq X$ and $q \notin Z$, so the hypothesis gives $b(X) \leq b(Z \cap X)$, and Lemma 4.5 gives $b(Z \cup X) \leq b(Z)$: the step costs no more than the original. After step $j - 1$ the closed set contains T_j with the same opened set, and the rest of the order applies. The search’s form follows by Lemma 4.1. $\square$

This theorem is not new. The hereditary premise is the commitment condition of Tamaki, and the proof is the proof of the commitment lemma, Kitsunai et al. (2016, Lemma 1, p. 142); Section 4.5 makes the correspondence exact. What we did not find in the papers we hold is the following cheap form of the test.

Theorem 4.7 (the repair is a matching condition). *Let $Y = X \setminus (S \cup \{q\})$, and join each $d \in Y$ to the stacks of $o(d, S) \subseteq o(q, S)$. Then q is hereditarily definite at S iff this bipartite graph has a matching with at least $\text{open}(q, S) - 1$ edges.*³⁵

³⁰Lean: `chuStuckey_theorem1_false, chuStuckey_theorem1_false_literal, chuThesis_theorem636_false.`

³¹Lean: `fink_theorem1_false.`

³²Lean: `IsHereditarilyDefinite, not_isHereditarilyDefinite_cex.`

³³Lean: `openStacks_submodular.`

³⁴Lean: `solvable_cl_insert_of_hereditarilyDefinite, searchSol_cl_insert_of_hereditarilyDefinite.`

³⁵Lean: `isHereditarilyDefinite_iff_hasDefiniteMatching.`

Proof. The sets of the repair are $B = S \cup D$, $D \subseteq Y$, and $b(X) \leq b(S \cup D)$ iff $|D| - |o(D)| \leq \delta$ with $\delta = |Y| + 1 - \text{open}(q, S)$, where $o(D)$ is the union of the $o(d, S)$. By the deficiency form of Hall's theorem the largest matching has size $|Y| - \max_D(|D| - |o(D)|)$, which is at least $|Y| - \delta = \text{open}(q, S) - 1$ exactly when the repair holds. The Lean proof reduces the deficiency form to Mathlib's Hall theorem with δ dummy stacks. $\square$

In words: the rule is safe exactly when all but one of the new stacks can be paired with a different freed customer who needed it. In the counterexample, closing 0 opens three new stacks and frees 3 and 4, both of whom needed only stack 0: one pairing where two are required. Every q with $\text{open}(q, S) \leq 1$ passes at once, so the published rule is right there.³⁶

4.3 The other rules

The subset rule. “If $o(c_i, S) \subseteq o(c_j, S)$ and $i < j \dots$ move j can be removed” (Chu and Stuckey, 2009, §2, p. 5). As coded, d dominates r when $o(d, S) \subseteq o(r, S)$ and the inclusion is strict or $d < r$; every dominated playable r is dropped, with a fallback that keeps P if all would go. The covering is sound for every set S : if $o(d, S) \subseteq o(r, S)$ and r is playable, then d is playable and $\text{Sol}_k(S \cdot r) \Rightarrow \text{Sol}_k(S \cdot d)$, because d is free after r and closing d then r reaches the same state at no greater cost. With the index tie-break, domination is a strict order, so an undominated move with a solution survives, and the fallback never fires when a solution exists.³⁷ The tie-break is necessary: without it, two customers with equal new stacks dominate each other and both go, which loses a solution on a 7-customer graph.³⁸

The better move. Chu and Stuckey (2009, §3.2, p. 6): “Suppose $S \mathrel{++} [q]$ and $S \mathrel{++} [r, q]$ are playable and $\text{close}(q, S \cup \{r\}) \geq \text{open}(q, S \cup \{r\})$ then if $U' = S \mathrel{++} [r] \mathrel{++} R$ is a solution there exists a solution $U = S \mathrel{++} [q] \mathrel{++} R'$.” Its premise is Theorem 1's premise for q at the child $\text{cl}(S \cup \{r\})$, together with playability of r then q , and its conclusion swaps q and r .³⁹ The swap is sound; the appeal to Theorem 1 is not. On the graph of Counterexample 4.4 at the root with $k = 6$, the definite move does not fire, customers 0 and 2 both survive the subset rule, and the premise holds for $r = 2$, $q = 0$, so the rule drops 2 citing 0. But $\{2\}$ has a solution and $\{0\}$ has none.⁴⁰ At that node another kept customer still has a solution, so this is a false link and not a lost node; no node where the better move loses the last solution has been found. Chu (2011, Thm 6.3.8, p. 143) states the better move with a different premise, $\text{close}(q, S) \geq \text{open}(q, S \cup \{r\})$, and proves it through Theorem 6.3.6. The same graph refutes it at $S = \{2\}$, $r = 3$, $q = 0$.⁴¹ The repair cites q for r only if q is hereditarily definite at $\text{cl}(S \cup \{r\})$, one matching at the child; then the swap argument goes through.⁴²

Two earlier faults of this rule in our own implementation are now proved in Lean too: a close count that included the customers r finishes on its own, and an order of the rules in which the subset rule could cite a customer the better move had already dropped, which loses a node through a cycle of true links.⁴³ The second is an instance of the general observation of Beck et al. (2025, Prop. 3, Example 6) that mutually dominating transitions must be tie-broken.

The old move. Chu and Stuckey (2009, Thm 3, p. 7) let a child inherit a sibling q whose subtree failed, provided that inserting q before each later step keeps that step playable. The rule is sound as coded: if

³⁶Lean: `isHereditarilyDefinite_of_openCount_le_one`.

³⁷Lean: `searchSol_cl_insert_of_newlyOpened_subset, subsetFilter_sound`.

³⁸Lean: `noTieBreak_counterexample`.

³⁹Lean: `isBetter_iff`.

⁴⁰Lean: `betterMove_counterexample, chuStuckey_theorem2_false`.

⁴¹Lean: `chuThesis_theorem638_false, chuThesis_theorem638_false_literal`.

⁴²Lean: `IsRepairedBetter, searchSol_cl_insert_of_repairedBetter`.

⁴³Lean: `bugA_counterexample, bugB_counterexample`.

$\text{cost}(S \cup \{q\}, c) \leq k$, then $\text{Sol}_k(\text{cl}(\{q\} \cup S \cdot c)) \Rightarrow \text{Sol}_k(S \cdot q)$, so a refuted sibling stays refuted along any path whose steps pass the test.⁴⁴ The test is necessary; the smallest graph that shows it is the path on five vertices.⁴⁵

The memo. A refuted state is recorded and refused when met again. Whether a state with $b(S) \leq k$ has a solution depends on the state alone, not on the path that reached it.⁴⁶ One implementation of ours declined to run the memo beside the old move, on the ground that a refutation obtained with inherited old moves depends on the ancestors. A node taken on its own, with an old move that is not in fact refuted, can indeed record a solvable state;⁴⁷ the next theorem shows that this never happens in a real run.

4.4 The repaired search is sound

Proposition 4.8. *At every state with the invariant, for every set Q of refuted old moves, the filter definite $\rightarrow$ subset $\rightarrow$ better with both repairs is node-sound.*⁴⁸

Proof. If the definite move fires, Theorem 4.6 applies. Otherwise the subset rule keeps a survivor with a solution; let w be the earliest. If the better move dropped w , it cited an earlier survivor that the repaired argument proves solvable, contradicting the choice of w . $\square$

Theorem 4.9 (runs are sound). *Let the filter be node-sound and return only playable customers. If at the start of a run the invariant holds, every memo entry is a genuine refutation, and every old move is refuted, then the run leaves only genuine memo entries, and if it answers false at S then $\neg \text{Sol}_k(S)$. This holds for any child order and any memo policy, with the old move on or off.*⁴⁹

The proof is an induction over the run, in the order in which subtrees complete: every old move a node holds is a sibling subtree that completed with *false* earlier in the same run, or an inherited one that passed the reinsertion test.

Theorem 4.10 (the revised search is sound). *Run the customer search with free moves, the memo, old moves and the filter definite $\rightarrow$ subset $\rightarrow$ better; the definite move firing only on a hereditarily definite customer and the better move citing only under the repaired premise. If it refutes k from the root, then no closing order costs at most k , $\text{pw}(G) > k - 1$, and on the MOSP graph of an instance M with a requirement, $Z(M) > k$.*⁵⁰

What the Lean model covers is the search as specified; that our C and Python implement it is checked, not proved. A transcription of the Lean filter agreed with the production filter on 16,244,090 node checks over graphs on 1 to 17 vertices; the matching agreed with the hereditary premise on 36,939,226 candidates; and a search written from the theorems agreed with the C, the Python and an exact oracle on 17,431,232 runs over 52,592 graphs on 1 to 17 vertices.⁵¹ Under the published rules the filter is not node-sound,⁵² and in a check of 570,206 runs with memo and old move it lost the last solution at a node in 58 runs, all on graphs built around Counterexample 4.4 and all at k equal to the optimum. Every one found a solution

⁴⁴Lean: searchSol_reinsert, not_searchSol_of_oldMove.

⁴⁵Lean: reinsert_needs_test.

⁴⁶Lean: solvable_iff_of_cl_eq.

⁴⁷Lean: exec_fake_oldMove.

⁴⁸Lean: repairedFullFilter_sound.

⁴⁹Lean: Exec.sound.

⁵⁰Lean: exec_repairedFullFilter_sound, exec_repairedFullFilter_pathwidth, exec_repairedFullFilter_mospValue.

⁵¹paper2/solver_fix.md, items 01–05, and paper2/revised_algorithm.md, section 4.4.4, with the regenerate commands.

⁵²Lean: not_codeFilterSound_cexGraph.

Table 4.1: The customer search in the vocabulary of exact pathwidth algorithms.

Customer search	Layout	Lean
closing order from $\emptyset$ open stacks $b(T)$	vertex sequence, read forwards border size $d(T)$	<code>orderOfLayout</code> <code>openStacks_eq_card_</code> <code>boundary</code>
$\text{cost}(T, c)$	$d(T \cup \{c\}) + 1$	<code>stepCost_eq_card_</code> <code>boundary_add_one</code>
$\text{Sol}_k(\emptyset)$	$\text{vs}(G) \leq k - 1$	<code>searchSol_empty_iff_</code> <code>pathwidth_add_one_le</code>
free closure $\text{cl}(T)$	full set of T (Suchan and Villanger, 2009, p. 328)	<code>mem_cl_iff</code>
memo	failure table	—
definite move, repaired	commitment to the child	<code>isHereditarilyDefinite_</code> <code>iff_isCommittable</code>

through another branch and answered correctly. No whole instance on which the published rules answer *unsat* wrongly is known.

4.5 The search in pathwidth language

For a set T of vertices write $N(T)$ for its *border*, the vertices outside T with a neighbour inside, and $d(T) = |N(T)|$, in the notation of Kobayashi et al. (2014, p. 390). A vertex sequence is w -feasible when every prefix has $d \leq w$, and $\text{vs}(G)$ is the least w for which some permutation is w -feasible. Table 4.1 translates.

Kobayashi et al. (2014, Algorithm 1, p. 393) decide whether a vertex set extends to a w -feasible permutation: replace the set by a committed extension, answer from a failure table or at V , recurse on each feasible one-vertex extension, record failures. The customer search is this loop with $w = k - 1$: the free closure and the definite move play the part of the commitment, the memo is the failure table, and the cost cut is the feasibility test. Coudert et al. (2014, §3.2) has the same three parts in a branch and bound. Neither line cites the other in the papers we hold: Chu and Stuckey (2009) predates the commitments and both 2014 papers, and none of the pathwidth papers cites it.

Commitments. Kitsunai et al. (2016, p. 142), stating Tamaki’s notion, call an extension τ of σ *commit-table* when $d(X') \geq d(V(\tau))$ for every X' with $V(\sigma) \subseteq X' \subseteq V(\tau)$; their Lemma 1 says that if σ extends to a feasible permutation, so does τ .

Lemma 4.11. *For $q \notin S$ and $X = \text{cl}(S \cup \{q\})$: the published premise holds iff $d(X) \leq d(S)$, the commitment condition at the single set $X' = S$; and the repaired premise holds iff X is a commitment from S , the condition at every X' between them.*⁵³

So Theorem 4.6 is the commitment lemma at the child, and the published Theorem 1 is a commitment whose interior condition was dropped. In Counterexample 4.4 the endpoint test passes with $d(X) = 3 = d(S)$, while the interior set $\{2, 3, 4\}$ has border 2.⁵⁴ The published premise is not wasted: by Kitsunai et al. (2016, Lemma 10, Cor. 2, pp. 148–149) it guarantees a commitment to the least-border set between S and X , which need not contain q .⁵⁵ In the counterexample that set is $\{2, 3, 4\}$.⁵⁶ Committing there when the matching fails is a second repair that gives up no pruning; we have not built it.

⁵³Lean: `isDefinite_iff_endpoint, isHereditarilyDefinite_iff_isCommittable`.

⁵⁴Lean: `cex_isDefinite_not_isCommittable`.

⁵⁵Lean: `exists_isCommittable_of_isDefinite`.

⁵⁶Lean: `cex_isCommittable_234`.

Table 4.2: Each rule of the customer search against the exact-pathwidth literature.

Rule	Published counterpart	Status
free move	full sets (Suchan and Villanger, 2009); Kitsunai et al. (2016, Prop. 3)	known
definite move, open ≤ 1	depth-one commitments; greedy step; push rule	known
definite move, published	–	false (Cex. 4.4)
definite move, repaired	commitment lemma (Kitsunai et al., 2016, Lemma 1)	known; matching test (Thm 4.7) not found
subset rule	none found	proved here
better move, published	none found	false
better move, repaired	none found	proved here
old move	none found	proved here
memo	subset DP (Bodlaender et al., 2012); prefix and failure tables	known; with the old move, proved here (Thm 4.9)

A commitment of depth one, S to $S \cup \{q\}$, holds iff $\text{open}(q, S) \leq 1$.⁵⁷ The greedy step of Coudert et al. (2014, Lemma 3, p. 49) is exactly that condition,⁵⁸ and the push rule of Suchan and Villanger (2009, Rule 1, p. 330) is a case of it. So every published pathwidth rule of this kind lies in the region $\text{open} \leq 1$, where the published definite move is right; Kobayashi et al. (2014, p. 388, Table 2) report depth-one commitments as “extremely effective”. Chu and Stuckey’s Theorem 1 reaches into $\text{open} \geq 2$, and that is where it fails. For the cost of the test, Kitsunai et al. (2016, Cor. 2) find the least border between the ends of a given extension by a minimum s – t separator; Theorem 4.7 is that computation for the single target X , where it becomes a bipartite matching.

Table 4.2 sums up what is known, limited to the pathwidth papers we hold (Suchan and Villanger, 2009; Bodlaender et al., 2012; Coudert et al., 2014, 2016; Kobayashi et al., 2014; Kitsunai et al., 2016). “None found” is not a claim of priority.

Prior reports. We looked for any earlier report of the error. We listed the works citing Chu and Stuckey (2009), Chu’s thesis and its IJCAI 2013 abstract in OpenAlex, Semantic Scholar and OpenCitations: 141 works. Of the 38 that cite the CP paper we read 24 in full (63%), and of the 103 that cite only the thesis, 32 (31%).⁵⁹ No work read gives a counterexample, an erratum or a correction. One, Fink (2012), restates the definite move as true with a stronger premise, refuted above. Three use an implementation of the rules and report no failure; among them Frinhan et al. (2018) ran the original code to obtain the optima that Martin et al. (2022) reuse. We found no prior report of the error. The pathwidth literature has the correct form of the rule, as a commitment, and none of the pathwidth papers we hold mentions MOSP.

4.6 The revised algorithm in practice

The revised search changes two tests. The definite move fires on the first playable q that passes the published test and then the matching of Theorem 4.7; a q that fails the matching is passed over. The better move cites q for r only if the same matching succeeds at $\text{cl}(S \cup \{r\})$. The matching joins at most close -1 customers to open stacks and stops at open -1 edges. Since 1 October 2026 these are the default in both of our solvers,

⁵⁷Lean: `isCommittable_insert_iff`.

⁵⁸Lean: `isGreedyStep_iff`.

⁵⁹`python3 -m paper2.prior_art_sweep`; the table of citers is in `paper2/prior_art_counterexample.md`.

Table 4.3: The cost of the repair: nodes under the published and the repaired rules, in paired runs of the same decision in one worker. Totals over the pairs that finished under both settings; a censored pair says nothing about cost. MOSP rows refute the optimum minus one; the graph row is whole descents at 120 s each.

Where	Vertices	Pairs finished	Nodes, published	Nodes, repaired	Ratio	Worst pair
MOSP corpus, default	9–40	6,135 of 6,135	228,147	228,154	1.00003	1.009
MOSP corpus, csearch	9–40	6,135 of 6,135	202,962	202,971	1.00004	1.040
Chu & Stuckey classes	50–100	118 of 125	1,358,096,671	1,362,205,210	1.00303	1.013
Chu & Stuckey classes, cap 2×10^8 nodes	125	11 of 23	90,969,508	90,961,419	0.99991	1.000
Graph pathwidth, whole descents	4–2,916	731 of 880	2,977,849,807	3,038,670,717	1.020	–

Table 4.4: Re-certification of the hardest corpus values under the repaired rules, by a split of the root into independent tasks. k : the value minus one, which must be refuted. For an instance in flight, the nodes and core-hours are those of the finished tasks so far.

Instance	Value	k	State	Tasks refuted	Nodes	Core-hours
Random-125-125-4-1_0	57	56	refuted	669	4.97×10^{10}	18.1
Random-125-125-2-4_0	24	23	refuted	1,686	4.98×10^{11}	154.1
Random-125-125-4-5_0	46	45	refuted	616	7.96×10^{10}	26.1
Random-125-125-4-2_0	57	56	refuted	842	7.64×10^{10}	30.2
Random-125-125-2-1_0	24	23	in flight	197	8.24×10^{10}	40.7
Random-125-125-2-5_0	20	19	in flight	117	5.27×10^{10}	42.5
Random-125-125-4-4_0	51	50	in flight	578	1.06×10^{11}	61.8

the MOSP customer search and the graph pathwidth solver, in C and in Python; the published rules remain behind a flag.

Table 4.3 gives the cost in nodes. The repair changes tree sizes by at most a few percent and never changed an answer. Per node it costs about 2% in the MOSP C engine, measured on six hard instances at 75–125 customers. The published test passes and the matching then fails on 0.07–0.27% of definite-move candidates, a rate that rises with size.

Re-certification. Of the 6,374 certified optima in our MOSP corpus, 6,259 do not depend on the customer search: they rest on an exact subset-lattice computation, a checked DRAT refutation, an earlier SAT search, or a lower bound equal to the value. The other 115 rested on the customer search under the published rules. The repaired solver has re-refuted the value minus one on 112 of them, and no value changed. Table 4.4 gives the seven hardest, 125-customer instances at densities 2 and 4, which needed a split of the root over many cores; three are still running at the time of writing, and keep their values as verified upper bounds until they finish.⁶⁰

Certificates. A refutation by the search is a claim about a tree of 10^{11} nodes at 125 customers, and nothing in Theorem 4.10 lets a reader check that the code ran the search it models. So the search emits a certificate: the tree, every pruning step with its rule and witness, and for each definite and better step the matching of Theorem 4.7. An independent checker, which shares no code with the search, recomputes every premise from the matrix, verifies each matching (distinct customers, distinct stacks, each stack newly opened by its customer, at least open – 1 edges), requires every chain of covers to end at a child or an old move with no cycle, and replays the tree. Each accepted link is a Lean theorem; that an acyclic covering by such links is sound is a short paper argument, not yet formal. Table 4.5 gives the results on the corpus at 9–75 customers.

⁶⁰`python -m paper2.solver_fix_split --summary`; the split is proved sound in `Search/Split.lean`.

Table 4.5: Certificates for the repaired rules on the certified corpus at 9–75 customers, refuting the optimum minus one, 60 s emission budget. *Unknown*: emission did not finish. Below 41 customers the default rows match `csearch` and are omitted.

Customers	Config.	Inst.	Verified	Rejected	Unknown	Definite	Better	Matching edges	gz bytes (median)	Check ms (median)
9–10	<code>csearch</code>	1,614	1,614	0	0	307	92	32	332	0.04
11–20	<code>csearch</code>	2,508	2,508	0	0	2,657	1,171	658	356	0.08
21–30	<code>csearch</code>	1,816	1,816	0	0	14,796	6,148	5,675	414	0.24
31–40	<code>csearch</code>	197	197	0	0	46,337	18,176	17,150	650	1.27
41–50	<code>csearch</code>	120	120	0	0	530,424	120,302	152,597	12,045	33.24
51–60	<code>csearch</code>	1	1	0	0	5,155	2,709	1,885	126,508	278.86
61–75	<code>csearch</code>	30	19	0	11	103,095	31,404	37,261	26,516	143.70
41–50	<code>default</code>	120	120	0	0	748,952	0	191,780	12,665	37.01
51–60	<code>default</code>	1	1	0	0	24,685	0	8,797	661,951	1198.43
61–75	<code>default</code>	30	20	0	10	312,897	0	110,118	34,162	187.03

Every refutation emitted within 60 s verifies, 6,275 of 6,286 under the `csearch` configuration, and none is rejected; the 35.5 MB of certificates check in 147 s. On the graph of Counterexample 4.4 rooted at $\{2\}$, the published rules emit an exhaustive tree that claims no solution; a checker of the published premises accepts it, and ours rejects it, even with a largest matching attached (one edge where two are needed).

4.7 A dataset with certified pathwidth

Section 3 makes one certified pathwidth an exact answer to every problem in Table 1’s exact core: pathwidth, vertex separation and the vertex separator game take the width w ; MOSP, gate matrix layout with multiple folding, one-dimensional logic, interval thickness, narrowness and node search take $w + 1$. We therefore built one dataset from every benchmark collection of those problems that we could obtain, and the MOSP corpus. The unit is the isomorphism class of the graph, the MOSP graph for a matrix instance: two instances with isomorphic graphs have the same value under every problem, so the dataset holds one record per class, with its members and their isomorphism maps. Each record carries the graph, the width, its provenance (*refutation*, *bound* or *upper bound*), a witness layout of exactly that width and, for a bound, a minor certificate that replays.

Table 4.6 gives the counts. Twenty collections, 21,754 instance files, reduce to 17,714 classes: the MOSP collections (Smith and Gent, 2005; Faggioli and Bentivoglio, 1998; Chu and Stuckey, 2009; de Carvalho and Soma, 2015; Frinhani et al., 2018), eleven VLSI gate matrix circuits (de Oliveira and Lorena, 2002), the VSPLIB sets (Duarte et al., 2012) and the Small graphs (Martí et al., 2008), the Rome graphs (Di Battista et al., 1997), and treewidth collections used as pathwidth benchmarks (van den Broek and Bodlaender, n.d.; Larisch and Salfelder, n.d.; Dell et al., 2017, 2018). Of the classes, 16,087 carry a certified pathwidth with a witness layout and 862 a witnessed upper bound. An independent checker, sharing no code with the solvers, re-reads all 20,987 member files with its own readers, maps each onto its record by isomorphism, re-measures every witness and replays every bound certificate, and reports no failure on the 16,949 records.⁶¹

Against published values: all 84 Small graphs agree with the pathwidths of Mallach (2018, Tables 4–5), and ten of the eleven VLSI circuits are certified at their published best-known track counts (de Oliveira and Lorena, 2002, Table I); the eleventh, W4 with 202 nets, is open at 28 against 27. One caution about MOSP benchmark files emerged: the same public repository ships the Carvalho and Soma instances as customers by patterns and the Chu and Stuckey instances beside them as patterns by customers, and only published densities and values tell them apart. Read the wrong way round, an instance certifies at a different value.

What is left is priced. One more budget step for every open class that fits our engine (1,024 vertices per

⁶¹`python paper2/dataset_check.py --sources; builder python -m paper2.dataset.`

Table 4.6: The dataset by collection. *Classes*: isomorphism classes within the collection (nauty certificates; 78 graphs above 5,000 vertices by file content only). *Owned*: classes whose first member is in this collection, so the column sums to the dataset’s class count. The provenance columns count owned classes with a witness layout.

Collection	Files	Classes	Owned	Refutation	Bound	Upper bound	No witnessed value
MOSP: Challenge 2005	5,806	3,169	3,169	3,168	1	0	0
MOSP: Challenge 2005 (second copy)	46	46	0	0	0	0	0
MOSP: Chu & Stuckey	200	200	200	198	0	2	0
MOSP: Faggioli & Bentivoglio	300	287	274	274	0	0	0
MOSP: SCOOP	24	24	24	24	0	0	0
VLSI gate matrix circuits	11	11	11	5	5	1	0
VSPLIB trees	50	28	28	28	0	0	0
VSPLIB grids	50	50	50	9	0	19	22
VSPLIB Harwell-Boeing	73	72	72	32	7	33	0
Small (Marti et al. 2008)	84	84	84	32	52	0	0
Rome graphs	11,534	11,199	11,199	8,126	2,746	326	1
TreewidthLIB colouring	58	58	58	18	13	27	0
freetdi named graphs	150	146	143	97	22	23	1
freetdi control-flow graphs	1,817	1,070	1,064	398	661	4	1
MOSP: Frinhani et al. 2018, large	610	610	610	0	0	0	610
PACE 2016	291	288	112	37	18	51	6
PACE 2017 exact	200	199	193	61	7	115	10
PACE 2017 heuristic	200	197	173	11	3	45	114
PACE 2017 bonus	100	100	100	23	0	77	0
MOSP: Carvalho & Soma 2015	150	150	150	11	0	139	0
All	21,754		17,714	12,552	3,535	862	765

component) costs at most 500 core-hours, and on the Rome graphs each such step has closed about 40% of what it attempted. The largest MOSP instances, 400 to 1,000 customers, and the 154 classes above 1,024 vertices are beyond an exact search. Refutation-certified records rest on the repaired search; for those within reach of the certificate emitter, the proof object above is the route by which a third party can check them without our code.

5 Closing

Placeholder. Section 5 is written last, once Sections 1–4 are final (`paper2/plan.md`, section 5). It is to say what the complex is, what is proved and where, and what remains open: the missing Table 1 reference (Kashiwabara and Fujisawa 1979); the Table 1 claims that turned out false or only bands; LaPaugh’s `es = pes`, stated and not formalised; the three re-certifications still running; the certificate’s acceptance condition, argued on paper and not in Lean; and the gap between pathwidth and treewidth on trees of cliques, where no degree or clique bound can close it.

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